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Youth engagement in community renewable energy in northern and Indigenous communities: Lessons from Peter Ballantyne Cree Nation and Frog Lake First Nation, Canada

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ABSTRACT

Transitioning from centralized, fossil fuel-based energy systems to decentralized renewable energy is critical for addressing climate change and ensuring energy security. Youth are key actors in the global energy transition, but there is limited understanding of the role of youth as energy citizens in rural and remote communities – especially Indigenous communities. This study examines the role of Indigenous youth in community renewable energy transition, drawing on semi-structured interviews with community members from Peter Ballantyne Cree Nation in Saskatchewan and Frog Lake First Nation, Alberta, Canada. Results show that notwithstanding the rise of youth engagement in the international energy space, the role of Indigenous youth participation in community energy transition is limited. Facilitating Indigenous youth engagement in local energy transitions requires intentionality, including formal means and opportunities to engage in energy governance; improved understanding of youths' envisioned social value of energy; addressing many of the enduring social challenges that disproportionately impact Indigenous youth; and integrating Indigenous culture and land-based learning in community energy programs and educational initiatives.

1. Introduction

Energy transitions are tightly coupled with responses to the global climate crisis [1,2], but energy transitions in many rural and remote regions are about meeting basic human needs and supporting social and economic livelihoods [3]. Around 10 % of the worlds' population still lacks access to electricity, and over 3-billion people still depend on unclean sources of energy for heating and cooking [4]. Many rural and remote communities, including many Indigenous communities across Canada, do not have access to clean and reliable energy and depend on aging energy infrastructure or off-grid diesel generators [5–7]. These communities experience energy injustices, including high home energy bills, vulnerable energy infrastructure, and constraints to affordable energy supply for investing in healthcare, youth education, and community economic development opportunities [8–10].

Community renewable energy (CRE) projects [17] are often seen as a solution to energy insecurity in rural and remote regions [18–21]. CRE can provide accessible, affordable, and sustainable energy, thus eliminating energy poverty [23–25], improving socio-economic opportunity [26], enabling energy self-sufficiency [27–29], and in the Canadian context reconciling relationships with Indigenous communities [30,31]. However, the implementation of CRE projects in rural and remote communities can face many challenges, from limited energy infrastructure and financial capacity [32,33], to policies that constrain energy transition [34,35] or that *impose* energy solutions on communities without consideration of local community capacity, values, and priorities [15,36,37]. The immediate and long-term success of CRE initiatives depends heavily on the engagement and support of energy citizens [11] in CRE planning, visioning, and implementation [38] – and especially youth participation and leadership [39]. Youth are essential and legitimate citizens in the process of a just and inclusive energy transition [38,43,95]. Yet, whilst scholars have discussed the importance of youth in energy security [38,66], especially youth education and employment [54,55] and climate action [64,65], we are not aware of any empirical studies that specifically explore youth as energy citizens, especially among marginalized and Indigenous populations.

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1.1. Energy citizenship

There is a plethora of definitions of energy citizenship, from the role of individuals in acting responsibly about energy consumption [8,12], to the engagement of individuals in energy arenas through democratic and civil opportunities at local to the supra-national scales [4,5]. Debourdeau et al. [11] explore the diverse use and interpretation of energy citizenship in recent scholarship, and posit that energy citizenship is more appropriately conceptualized as a multi-dimensional construct, encompassing both agency attributes and outcome orientations. Agency attributes refer to whom energy citizenship is ascribed, whether individuals or collectives, and thus hinges in part on opportunities or pathways for engagement, the knowledge and motivation to engage, and the capacity to engage [12]. Outcome orientations refer to how energy futures are imagined, and the reformative or transformative change that is desired by energy citizens [11], such as the perceived benefits that may accrue to local communities, economies, or wellbeing. Both agency and outcome dimensions are important to exploring energy citizenship and to the pursuit of energy transitions, policies, or technologies that are considered appropriate or legitimate [6,7]. However, despite extant literature on energy citizenship, including the growing scholarship on the engagement of historically poor and marginalized populations [21], the socio-economic positioning of energy citizens is poorly understood [13] and there remains a need for critical discussion as to "who counts as a citizen" [22], especially among youth, and what is to be achieved via energy citizenship.

1.2. Youth as energy citizens in energy transitions

There is no single definition of what constitutes "youth", but youth are often characterized as individuals between 15 and 30 years of age [40–42]. The world youth report of the UN Department of Economic and Social Affairs notes that active engagement of youth is central to sustainable, inclusive, and more stable societies and to finding solutions for complex sustainability challenges [14] – including those associated with energy insecurity. Bastien & Holmarsdottir [43] emphasize the importance of youth-driven innovation for social change, and the UN [14] characterizes youth as a driving force for mobilizing political change. Other scholars have emphasized the important role of youth in energy transitions from the local to the global level [39], alongside the overall importance of youth to future policy, economic development, and community leadership [38,45]. Canada's first State of Youth Report [95], for example, characterizes youth as "uniquely well-positioned to inform an innovative and inclusive future for everyone" and as "key drivers of social and economic outcomes."

Indigenous youth are critical actors for the responsible building of sustainable, inclusive, and resilient energy futures in rural and remote regions [46]. The population of Indigenous youth in Canada is growing significantly faster than non-Indigenous youth [95]. The Seventh Generation Principle, an Indigenous concept focused on thinking seven generations forward,[1] emphasizes that the active participation of youth in any community initiative is crucial for long-term community futures [44,45]. Accordingly, Indigenous youth as energy citizens requires the attention of researchers. However, international scholarship reports that youth are often excluded from consultation and decision-making processes regarding energy development and operations [89], despite being significantly affected. Among rural communities in Indonesia, Ratriyana et al. [66] found that the values, norms, and beliefs upheld by community elders and leaders often diminish the voices and roles of youth, thus hindering their effective social or political engagement in CRE initiatives. Other researchers have stressed the dominance of men over youth (and women) in energy initiatives, describing local CRE projects an "exclusive club for men" [93].

Exacerbating these challenges is extensive evidence that rural and remote Indigenous communities confront a variety of injustices, such as social marginalization, enduring poverty, land rights loss, restricted resource access, and heightened susceptibility to the effects of climate change [47]. Despite efforts aimed at enhancing their well-being, many such communities globally still endure marginalization and prejudice, alongside enduring energy insecurity [38]. Development initiatives, including those focused on energy solutions, have typically been top-down [48], neglecting crucial aspects vital to the survival, identity, perspective, and lifestyle of communities – local knowledge systems and youth [49]. In the Canadian context, historical events have profoundly affected Indigenous peoples, with forced relocation onto reserves, for example, coupled with limited social and economic resources resulting in increased poverty and dependency on outside governments [50,51]. Compounded by the geography of remoteness, many Indigenous communities have lagged behind urban communities in terms of basic infrastructure – including energy infrastructure – contributing to social disruptions and constrained socio-economic opportunity [16,45,52]. The impacts of such are often disproportionately felt among youth [51], who are often underrepresented or not engaged in decisions about their future [53].

1.3. Research question

Failure to address the above challenges risks youth being left behind in energy transitions in rural and remote communities, in comparison to youth in more privileged urban and non-Indigenous settings. Jaradat et al. [39] reviewed the state of international scholarship on youth in energy transitions research and conclude that whilst youth participation, as energy citizens, in energy transitions has been addressed, the focus is largely on youth energy education [54–57], conservation behaviour [58–60], energy awareness [61–63], or climate advocacy [64,65]. Moreover, notwithstanding the recognized importance of youth engagement to sustainable community and energy futures [38,66,67], there has been relatively limited attention among energy transitions scholars to whether and how Indigenous youth help shape CRE futures and imaginaries at the local scale [39].

The purpose of this paper is to examine the nature and role of youth energy citizenship in CRE in northern Indigenous communities. Specifically, this research asks: how are Indigenous youth shaping their community's energy future(s), and what barriers and opportunities exist to youth engagement in CRE? This is done based on learning from the experiences of Indigenous communities in northern Saskatchewan and Alberta, Canada. The lessons emerging are of broader value for the energy scholarly community and those engaged in energy planning in other regions.

Specifically, our research makes an important contribution to current scholarship on youth energy citizenship, responding to calls for contextual and empirically based research that seeks to understand whether and how Indigenous youth shape community energy futures [39]. Scholars emphasize the power of youth as catalysts for energy transition and agents of change [66,96], suggesting that Indigenous youth should be equipped to lead CRE innovation, leveraging their "energy, talent and, eventually, expertise" to address the energy and broader social challenges that confront Indigenous communities [97]. However, our research illustrates that underlying social community challenges may inhibit youth active engagement and their potential to shape community energy futures. Limited access to education and resources, intergenerational tensions, systemic inequalities, and poverty within Indigenous communities can restrict youth participation in CRE initiatives or further socio-economic projects.

2. Study area and methods

To understand how youth are involved in shaping their community energy futures, our research focused on the lived experiences and

[1] https://theseventhgeneration.org/blog-the-seventh-generation-principle/

perspectives of youth and other community members from three Peter Ballantyne Cree Nation (PBCN) communities in northern Saskatchewan, Canada, and Frog Lake First Nation (FLFN) community northwest Alberta, Canada (Fig. 1). PBCN and FLFN are engaged in research partnerships with the Community Appropriate Energy Security (CASES) initiative at the University of Saskatchewan, or with CASES project partners including QUEST Canada – a registered Canadian non-profit focused on energy resilient communities.

2.1. Peter Ballantyne Cree Nation

Peter Ballantyne Cree Nation (PBCN) is an Asinīskāwiyiniwak First Nations band government, comprised of eight communities in northern Saskatchewan: Denare Beach (Amisk Lake), Deschambault Lake, Kinoosao, Pelican Narrows, Prince Albert (Kiskaciwan), Sandy Bay, Southend, and Sturgeon Landing. This research engaged the communities of Deschambault Lake, Pelican Narrows, Southend. The territory is historically inhabited by the Cree Nation. The Cree Nation administration consists of an elected Chief and Council, comprised of 14 members from PBCN communities. Each community maintains a band office. The total population of PBCN is 12,002 as of December 31, 2022 [68] – the most recent official population data available, and just under 30 % of the First Nation population in Saskatchewan is youth. The employment rate in PBCN is low with most occupations in government services (i.e. health and education), and in the resource sector including forestry and fishing. The unemployment rate in Pelican Narrows is 23.2 % [70], 22.6 % in Descambault Lake, and in Southend 21.4 % [72]. This is high compared to the province, which is 7 % [69]; however, the employment rate is just 19.1 % in Pelican Narrows [70], 22.9 % in Deschambault Lake, and 37.6 % in Southend [72], compared to the provincial rate of about 60 %. The median age in these communities ranges from 17.4 yrs. (Southend) to 20.6 yrs. (Pelican Narrows), compared to the provincial median age of 38.8 yrs. (Table 1).

Deschambault Lake, Pelican Narrows, and Southend are connected to the province's northern electrical grid, owned and operated by SaskPower, the province's electricity service provider. However, none of

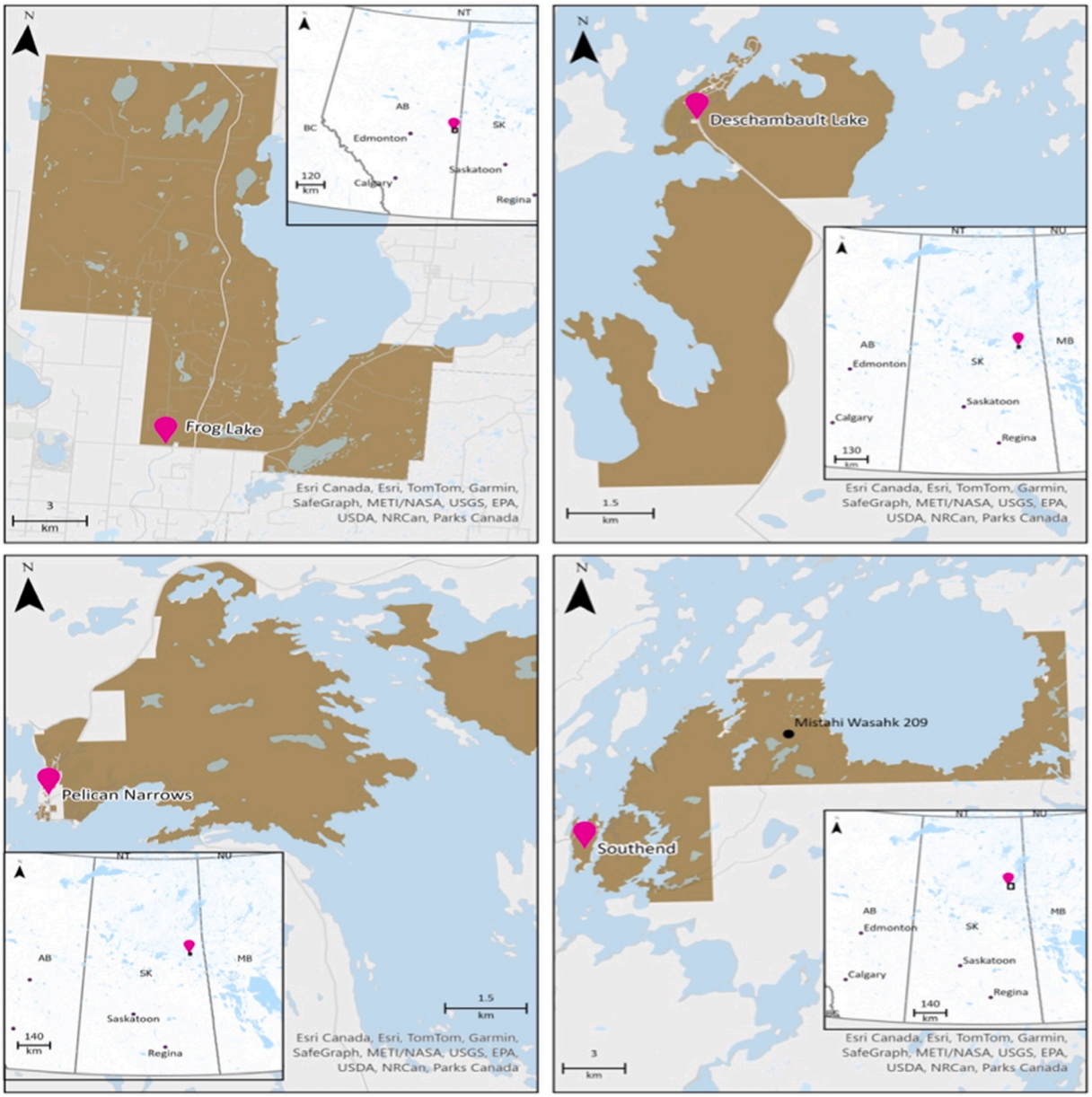

Fig. 1. Participating communities in Peter Ballantyne Cree Nation (Pelican Narrows, Southend, Deschambault Lake) and Frog Lake First Nation (Frog Lake). (Source: Map developed by Silas Asante, University of Saskatchewan.)

Table 1
PBCN study area community population profile (on reserve population).

	Population	Population change	Median age
Pelican Narrows[a]	1,837	-6%	20.6
Deschambault Lake[b]	840	-19.7%	20.2
Southend[c]	913	-0.4%	21.6

[a] Statistics Canada [70].
[b] Statistics Canada [71].
[c] Statistics Canada [72].

the communities are connected to the province's natural gas distribution network for heating. The communities have been characterized as energy insecure [73,74], owing to aging energy delivery infrastructure, frequent power outages caused by winter storms or seasonal forest fires, long service delays for power outage recovery, and a reliance on electricity for residential heating. Communities consume non-subsidized electricity to meet most of their energy needs, at a rate of $0.14895/KWh for households [75]. Electricity rates are the same throughout the province; however, with no connection to the natural gas network, coupled with colder winters in the North and larger than average household size owing to many multi-generational households, the costs of monthly energy are considerably higher than for the rest of the province [73,74,76]. All three study communities are engaged in some capacity in CRE initiatives, albeit at the very early stages and with a focus on baseline data collection on energy use, home energy efficiency, and technology or resource (e.g. biomass) assessments.

2.2. Frog Lake First Nation

Frog Lake First Nation (FLFN) is a Cree community in Alberta, Canada, approximately 80 km south of Cold Lake. The community is accessible year-round by paved secondary roads. Comprised of two reserves, Unipouheos and Puskiakiwenin, FLFN is governed by an elected Chief and Council who oversee community services, including employment and training, education, infrastructure, housing, health, and youth programs [77]. The population of Frog Lake is 1430 in 2016, of which 56 % are under the age of 24 and a median age of 22 yrs., compared to 36 yrs. for the province of Alberta [78]. Most housing is owned by the First Nation Band Council. The unemployment is approximately 44 % [78].

Located in the Alberta oil sands region, FLFN is heavily involved in the oil and gas sector. Recently, FLFN entered a partnership with a private sector company to construct a net-zero natural gas-powered electricity plant. FLFN is connected to the province's electrical grid, with ATCO-Electric, a public energy utility, being the distributor and retailer. Of the total 6,480,017 kWh of electricity that was consumed by the community in 2020, residential buildings consumed 67 % or 4,317,1118kWh. FLFN is currently engaged in feasibility studies for investment in solar energy, with an expressed commitment to improve energy efficiency and reduce reliance on oil and gas [79].

2.3. Methods

Interviews are commonly used by researchers when exploring social aspects of energy transitions [13,16,48,80], and in other contexts when attempting to reveal local, lived experiences, values, beliefs and motivations [81]. When working with Indigenous communities, interview processes are often more conversational in nature and offer a "culturally organic means to gather knowledge" [98]. In our research, whilst our conversations with community members about the role of youth in energy initiatives and futures were informal and occurred via both individual and group conversations, they were also guided by specific topics – an approach described as "research yarning" [16], whereby conversation is directed yet characterized by informality and reflexivity.

The guiding questions for our interviews were informed by Jaradat et al. [39] analysis of emergent themes and gaps in recent scholarship on youth as energy citizens, and by topics explored at recent international youth energy forums [65]. Interviews explored both agency and outcome dimensions of energy citizenship, specifically, youth perspectives of the community's energy future; the extent to which youth are involved in their community; and the perceived importance of youth involvement in charting the community's energy future alongside potential barriers. The conversational approach allowed for reflection by both the participant and interviewer, and for exploration of emerging questions and insights [83]. New or emerging topics were then explored in subsequent interviews [84].

Interviews were in-person and conducted in collaboration with larger community energy research initiatives by way of established partnership agreements or memoranda of understanding with the respective Indigenous Nations, as described below. Consent to be interviewed was obtained orally at the time of the interview and consent documented by the interviewee. Data collection and consent protocols were approved by the University of Saskatchewan Behavioural Research Ethics Board (Beh 1616) in December 2022.

Data collection at Deschambault Lake was conducted in collaboration with the CASES initiative, a research partnership to which PBCN is a signatory and provides shared oversight. Participants were identified by PBCN, thus ensuring community appropriate representation and participation. Interviews were conducted in group settings and individually and included Indigenous band members, business owners, educators, and other community members and elders. Only one youth participated at Deschambault Lake, notwithstanding the recruitment efforts of a local band employee and research liaison for the community.

Data collection in Frog Lake, Southend, and Pelican Narrows was conducted in collaboration with community energy mapping workshops that were led by QUEST and Co-Mapping Solutions, in collaboration with FLFN, PBCN, and CASES. The workshops were organized by the local community energy coordinators at Frog Lake and for Southend and Pelican Narrows. The Indigenous community's local community energy coordinators led the process of inviting participants to the workshops, through personal communications and community networks and using the communities Facebook pages. The workshops were for the purpose of community energy resource mapping. Thus, participants for this study were identified using intercept interviews [82], a direct method of interviewing and observing people while they were engaged in the mapping exercises. These workshops were not specifically targeting community youth and thus attracted participants from different age groups and demographics.

In total, across the four communities, 24 individuals participated in the research including 11 youth and 13 community members and leaders (Table 2). Three youth who were approached at the workshops declined to participate. The youth interviewed were between 18 and approximately 30 years of age, as per our research ethics approval (i.e. minimum of 18 yrs. of age) and based on discussion with community elders as to what constitutes "youth" in their community - i.e. up to 30 yrs. of age. Although we did not include school-age youth (i.e. < 18 yrs) in our sample, youth and non-youth participants provided insights to engagement in general in their community, and educators provided insights to school-based energy programming for youth. Non-youth interviewees, especially elders, provided insight to how they observe

Table 2
Study participants.

Community	Youth (18–30 yrs)	Community members and leaders (> 30 yrs)
Deschambault Lake	1	9
Southend	3	3
Pelican Narrows	3	1
Frog Lake	4	0
Total	**11**	**13**

youth engagement in their community and some of the enduring barriers to active youth involvement.

Interviews lasted between 45 and 90 min, were recorded, and transcribed verbatim immediately after the interview [85]. The interviewer also took notes, especially on new and recurring issues raised by participants. Through the transcription process, constant comparisons were possible as analysis began immediately following each interview and remained ongoing, so new concepts not initially discussed in the interview could be incorporated into subsequent interviews and tested in ongoing analysis [84]. Final interview transcripts were thematically coded with the assistance of NVIVO© v.12 software, through a process of coding-up or open coding to group and regroup interview data until common themes emerged We adopted Braun and Clarke's [86] framework for thematic analysis, comprised of data familiarization, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and presenting themes through a coherent narrative. Coding was done by the lead researcher to identify and group similar or complementary responses or points raised by interviewees, with codes and code groupings subsequently reviewed for validation independently by another researcher. The initial round of coding was simply to align the interview data with the three topics that guided the conversations – i.e. youth perspectives of the community's energy future; the extent to which youth are involved in their community; and the perceived importance of youth involvement and potential barriers. This was necessary owing to the conversational nature of the interviews and that not all conversations proceeded in the same order or with the same structure. This was followed by three additional rounds of coding-up to search for emergent themes and for cross-cutting themes. After four rounds of coding and grouping and re-grouping the interview data, four dominant themes emerged and are presented in the results section below. There was no intent to *compare* communities, especially given the small sample size and diversity of participants, but rather to gain insight from youth and other community members about youth engagement in shaping community energy futures and the common challenges and opportunities that emerge.

2.4. Limitations and potential biases

There are limitations to the research, namely those related to the challenges of engaging youth in northern communities and social boundaries. First, the interviewer is non-Indigenous, not from the North, and not from Canada. This may have caused some participants to be reluctant to engage, but project partners and local community energy coordinators helped facilitate introduction to the communities. Relying on community energy coordinators via local relationships influenced *who* participated in the research. Such community relationships are core to Indigenous epistemology serve to strengthen the credibility of the research design and its findings [98]. Whilst this may be seen through western science lenses as introducing potential bias, our approach provided a community-appropriate pathway to engage youth and community elders and to interact with knowledge holders in a way that may not have been possible with more rigid sampling designs. That said, we are cautious to not claim that the participants engaged in our study are *representative* of their communities. Second, the intent of this research was to focus solely on youth participants. We focused on youth above 18 yrs. of age, owing to the scope of our research ethics approvals and as discussed with community leadership. It proved challenging to identify a large sample of youth participants, even with the help of community members. The subsequent engagement of non-youth provided important perspectives to some of the challenges to youth engagement, especially perspectives from community elders and from those involved in youth programs or high schools. Third, the communities involved in this research are still positioned at the very early, pre-planning stages of renewable energy initiatives. Thus, there are no practical experiences for youth in the field of energy transition or opportunities to reflect on past engagement. This is the case for most northern and remote Indigenous communities across Canada. Thus, the focus of conversation was often on perspectives about new energy opportunities, how these opportunities fit into the community's knowledge systems and values, and how youth are engaged more broadly in their community.

3. Results

Results indicate many positive attributes for youth energy citizenship the study communities, due in part to the ambitions of these communities to overcome enduring socioeconomic hardships and the ambitions of youth not just to survive but to thrive. However, findings also indicate several obstacles to youth engagement, many of which are deeply embedded in the underlying community and social challenges and relationships. Four emergent themes resulted from our interviews that characterize the perspectives of PBCN and Frog Lake community members about youth engagement, namely: (i) Indigenous values and youth's perspectives on what is considered *community-appropriate* energy; (ii) the value-added that CRE projects are expected to offer communities; (iii) the extent to which youth are engaged in their community; and (iv) the underlying challenges to youth engagement in their community and in CRE transition planning. Given the current state of CRE in these communities, the conversations often focused on intent or ambitions, and youth engagement in community at large. Participants provided deep insights to factors and conditions regarding youth as active energy citizens that might influence the future of energy in these communities and other developmental endeavours.

3.1. Indigenous values and youth's perspectives on community-appropriate energy

Citizenship shapes and is shaped by values of the community. What is valued by any particular community has a profound impact on policy choices and how these may or may not reinforce cultural norms. A dominant theme emerging from the interviews was the importance of Indigenous values and how this underscores community way of life – not only energy security. Participants explained that land and nature, the source of living and thriving, are embraced by Indigenous people; every aspect of life, including energy, must be extracted from the land. This perspective was also evident during interviews with some community youth, as one youth participant described: "*everything we do in our culture is all from the land; everything we use…when we use our medicines…when we go and get our food, it's in the bush*". Participants also expressed their relationship with land and their responsibility to preserve it for future generations by balanced and clean consumption – including energy generation and use. A community leader elaborated: *"we were put here to preserve the land…. we take only for what we need, do as little damage as we can possibly,*" with another adding "*sustainability is not foreign to us [we] just call it different words."*

Interview participants, both youth and others, noted the alignment of CRE with Indigenous culture and way of living – energy that is rooted in self-sufficiency and ensures the preservation of land for future generations, such as solar, wind, and biomass. As one youth participant described, renewable energy is a natural source of energy that can sustain their community efficiently without relying on resources from outside the community. The youth went on to explain that "*if we do these clean energy projects, we're gonna be able to be self-sustaining and self-efficient with our land."* Youth are increasingly aware of the adverse impacts of fossil fuels for energy production, whether from oil sands in Alberta or coal-fired electricity generation in Saskatchewan. As one youth in Frog Lake commented, CRE is a means to protect the land and *"not be using it [land] in a negative way, like oil sands,"* but rather sustaining the resources for the next generation. The participant went on to express that "*with these clean energy projects, we're gonna be able to protect our land for generations of generations to come*".

Finally, participants emphasized the importance of cultural learning, including language and traditional knowledge, from one generation to

the next, in shaping how youth see and position themselves as actors in their community and thus its energy future. That said, owing to the traumas of oppression experienced by the older generations who went to residential schools, one youth participant expressed concern that "*the ability for elders to play a major impact within those families has been lost.*" The result was said to be an increasing sense of detachment among youth from their own culture and community, with potentially negative implications for whether or how strongly youth align their energy futures with Indigenous values.

3.2. Value-added expected from energy projects

Energy citizenship includes reflection and community discourse on how energy projects can enhance well-being of the community. Participants identified several social values and opportunities anticipated from the introduction of renewable energy projects in their communities. Economic revenues emerged as one of the most tangible values and benefits, which was identified as especially important given the current burden of high energy costs in communities alongside other underlying economic struggles such as youth unemployment. Participants especially emphasized the importance of not only having renewable energy projects in their community but also owning those energy projects. For example, as one youth community member explained, owning energy projects would mean that the community could be energy self-sufficient and "*not depending on [the] power company*" [i.e., the Crown energy utility]. The participant went on to explain that owning CRE projects would mean that the community "*can actually generate revenue 'cause we'd be paying towards the band [Band Council] instead of an external company*." In doing so, this was seen as an opportunity to move toward energy self-sufficiency and to generate economic returns to reinvest in the community and to further advance local energy projects.

For many youth participants, being energy self-sufficient was seen as an opportunity for new local employment, education and training, population retention, and being able to build a future in their community. Youth expressed enthusiasm about the potential for local renewable energy projects, based on expectations about new education and training programs in their community, and the possibility for post-secondary training in renewable energy technologies and skills that could lead to employment with the government, their community, or with private sector organizations. As one youth participant expressed, "*We could establish mentorship programs, apprentice programs, you know, have kids come and look at the factories and how it's done*." Some youth participants offered even larger aspirations for renewable energy, describing ambitions for youth to be the next generation of renewable energy leaders and to expand the industry of renewable energy projects into other communities. For example, one youth participant explained: "*What we could do…is be a leader in the industry, especially in First Nation communities and kind of create those blueprints for other communities*".

To achieve the full value of renewable energy, all participants highlighted the significance of youth engagement in energy planning and in projects for the future of their community, noting that youth comprise the majority of the population in the North. This means that youth must be more involved in both planning and executing local energy projects to ensure their long-term acceptability, viability, stability, and to fully recognize the local value added of renewable energy projects in the community. As expressed by one community leader, "*having youth involved in everything around is good,*" explaining that "*when something else comes around… five, six years in the future, that group [youth] will be around, just take part in it, and guide the next group*." Several participants, across all communities, noted this multi-generational role for youth in the future of CRE; however, they also cautioned that achieving this outcome requires empowering youth now, with knowledge and training, and engaging them in energy conversations and visioning. As expressed by one community member: "*they're [youth] the future, right; so, if you teach them, they'll carry on with hopefully better… projects that are clean energy*."

3.3. How youth are engaged

At the core of energy citizenship is participation in the body politic of the community. Notwithstanding the importance interviewees placed on youth in Indigenous communities and culture, coupled with the potential value that energy self-sufficiency provides for youth, participants indicated that youth engagement in their community, in general, is limited. This was described as problematic for sustainable energy futures, in that youth are seen as important actors and enablers of energy transitions. For example, one community leader explained that youth are not engaged in energy problems or initiatives and currently have no responsibilities or expectations placed on them to become engaged and, as a result, "*they* [youth] *don't think about things like that [i.e. energy]*". Another participant emphasized the challenges to bringing the community, especially youth, together to plan for energy futures, noting: "*This is the tough part here, …bringing the nation together, finding that unity.*"

Given the limited CRE activity currently in the study communities, participants expanded the conversation to youth engagement in the community at large. Youth described their own engagement in community initiatives and visioning as *"poor and in short term."* Participants spoke about short-term employment opportunities, engagement on youth school councils, sports activities, community tree planting, and participating in traditional ceremonies. The band offices in both regions (PBCN and FLFN) sometimes provide job opportunities for youth in fields such as education, community services, and health care; however, participants indicated that youth unemployment is high – up to 30 % in PBCN and over 40 % in FLFN, leading to an increased youth dependence on government social assistance programs once school is completed, or even out-migration. Whilst a community leader in one study community emphasized that schools have a pivotal role in youth empowerment, and that band council and leadership can "*go to the school and do our informational program,*" several youth participants noted that such initiatives were not effective at engaging students or building leadership skills and connections with Indigenous and other agencies or businesses. As one youth expressed: "*There used to be an intro to agency meetings where they, the youth, could be part of, but that doesn't happen anymore.*"

Interviewees noted that some youths are trying to make a difference in their community, noting that *"we have a lot of ceremonies that we're trying to keep alive in our community and you know it comes from the youth."* The challenge, explained a community leader, is "*how to engage them [youth] …to increase their motivation to work in the community for the best of the community*", including in energy initiatives. We did, however, speak with one youth from a study community who started his own home energy retrofitting business, expanding an existing construction business. The business is focused on community building energy efficiency, such as adding energy retrofits to new or renovated homes. In addition, through this business he teaches and trains youth about energy efficiency and renewable energy technologies.

However, results indicate that youth currently have limited formal voice in planning for development projects or for sharing their perspectives with the Indigenous band councils or boards. Youth might have the right to participate on various committees, as one community member explained: "*There are youth committees … because policy states or the board has to have that.*" But the interviewee went on to note that there is a prominent absence of youth contributions and "*there's no real input.*" Some youth attributed their lack of representation in community planning and decision-making process to community leaders, with one youth indicating that leaders "*want to engage youth, but they don't apply it well.*" Other youth participants claim that youth are "*oppressed*" and are unaware that their voices matter; "*they don't understand that their opinion means something, or it's valued.*" Another youth participant went on to explain that youth don't understand the importance of their voice in community leadership: *"They don't understand that because they're not taught that…they're not gonna speak up during engagements because they don't realize that they have the voice."* Youth participants across the study

communities have a strong desire to be effective actors in community planning and development, including community energy futures, aspiring to have a meaningful voice. As one youth participant, who is also a community energy coordinator, described: "*I definitely think that's something [my community] struggles with, is not having enough youth engaged in community planning…I definitely think that's something that the [community] needs to work on.*"

3.4. Underlying conditions as barriers to youth engagement

Building on conversations about how youth as energy citizens are engaged in their community, three underlying conditions emerged from the interview results that reveal barriers to youth agency and, thus, may help explain the constraints to youth as local agents of change in community energy transitions.

3.4.1. Limited community infrastructure to mobilize youth

Full energy citizenship requires the opportunity to participate, as well as efficacy of that participation. Results indicate limited social and community infrastructure to engage youth, to foster a sense of community, and thus to motivate youth participation in community initiatives. Through our conversations, it was clear that in some communities the high cost of operating community infrastructure is seen as an obstacle to youth engagement; often, communities can barely afford to meet essential infrastructure needs. Specifically, participants indicated that the cost of heating and powering community buildings in the North limits the resources available to invest in youth, via youth centers, extracurricular activities, and recreational facilities. The focus on meeting basic community needs, explained one community councillor, "*affects them [youth] directly.*" A specific example noted in one community was the recent closure of the community's recreational facility. An interview reported that the building required more than $100,000 CAD annually simply to cover utility bills. Notwithstanding the recognized importance of such facilities to youth for combating isolation, facilitating a sense of belonging, and for motivating youth to be engaged in the community, they are "*very limited*" in the North *"and that's in correlation with the lack of funding that's in the community*." For youth to become engaged in broader initiatives like discussions about community energy futures, youth need to be active and inspired instead of *"staying home…couch surfing*." This requires that communities offer the places and means for youth to gather and engage with community; something that one community member said is "*almost impossible unless you have other [external] organizations pitching in to help.*" There is external government support for youth programs, but the focus is on the more critical needs related to health and social services instead of other forms of community infrastructure for youth.

3.4.2. Narrowly scoped educational programming

Citizenship requires civic literacy; energy citizenship requires energy literacy. A second condition concerns the nature of youth education and educational programming in the study communities. In general, according to study participants, educational curricula focus largely on social sciences with much less attention to environment and sustainability sciences or technical skills, which some interviewees said is a main reason for youth's lack of awareness of energy problems in their communities. Participants commented that the educational system in the North is designed to prepare youth for limited careers, all of which fall under social services: "*targeting a certain career path, more of a social sciences sort of school… putting kids through school to become teachers or nurses.*" Participants thus emphasized the need to transform curriculum to be more comprehensive and inspiring, where youth can learn more diverse subjects and choose more high-profile careers for the interest of the community's future, especially in energy and engineering. For example, a former teacher suggested that school programmes need to be designed to address energy by teaching practical knowledge whereby youth can learn in more creative ways. The individual went on to explain that by *"developing programs specifically targeting energy, the energy sector, they [youth] would become more aware of energy itself, energy security, and what that entails"*. However, notwithstanding the importance of technical and science-based education to training youth as energy citizens, participants emphasized the need to ensure that energy education in the communities taught "*from a First Nations perspective*" – a perspective that is largely absent from curriculum currently developed and managed by provincial governments. As one interview summarized, when education and programming *"comes from the community, from the people's culture, [and] traditions"* it can shape youth identity and empower them to be come active community agents of change.

3.4.3. Enduring social-wellness challenges

Energy citizenship requires political capacity. Many youth in the study communities are confronted with far more critical day-to-day challenges than those associated with their energy future. Interviewees commented on the challenges that Indigenous youth are facing, from remoteness and isolation to violence and generational substance abuse. These challenges prevent youth from being active members of their community, and from taking on leadership roles. One youth interviewee explained the feelings of isolation and lack of opportunity, noting that young people can be easily driven to escape reality by turning to "*the cycle for doing drugs and alcohol.*" Participants referred to past traumas, to "*youth who grew up in trauma-filled homes or drug and alcohol homes,*" and to the cultural and identify gap between the culture of Indigenous communities and broader Canadian society that can leave Indigenous youth feeling lost and struggling to find their place. As a youth participant expressed: "*A lot of young people are battling with anxiety; maybe it stems from not knowing where they're going, where they come from, and where they need to go*". This sentiment was reinforced by a community leader who described a lack of motivation among many youths, owing to a sense "*of hopelessness…like there's no future in anything.*" It's hard for youth to engage and be motivated to change their community energy trajectory, another participant explained, when there is limited social interaction and "*they don't have the confidence in themselves.*"

4. Discussion

This research set out based on the premise that youth energy citizenship is an important catalyst for successful sustainability transitions in general and a crucial factor for CRE in particular. This premise is based on assertions present in global societal discourses via many intergovernmental platforms, such as the United Nations and IRENA, and on scholarly literature focusing on the role of youth in the global climate movement [38,87,88]. Yet, at the same time, our understandings of youth energy citizenship at the local level remains limited [39,66,89]. Notwithstanding the importance of youth in energy transition and the increased engagement of youth in the national to international energy space, lessons from our research suggest a very different story in the context of northern Indigenous communities. Although there is strong youth interest in energy transitions and CRE in our study communities, there are significant barriers to youth as energy citizens.

Research focused on Indigenous CRE often emphasizes the legacy impacts of colonization and marginalization on local energy security, engagement, and future opportunities [31,36]. Our findings similarly reflect that some of the barriers faced by Indigenous youth as energy citizens are rooted in larger socio-cultural challenges and contexts, including the historical marginalization of Indigenous communities in general. However, in other instances our results reflect a misalignment between Indigenous youth values and their immediate expectations about CRE as a means for local social and economic opportunity versus the longer-term climate goals that are often espoused in recent youth energy scholarship [39]. Without appropriate pathways for Indigenous youth to participate and have efficacy in the energy futures of their

communities, it will likely be more difficult to ensure that CRE and energy transitions realize their fullest potential to align with community values and achieve full value-added girded by legitimacy of active participation of the entire body politic, including youth energy citizens. In the sections that follow, we identify four interrelated observations emerging from our research that are of importance to advancing and contextualizing scholarship on youth energy citizenship.

4.1. Mobilizing youth engagement in CRE

First, our results emphasize the need to identify and assess current programming and opportunities for youth engagement in CRE at the *local* level, including factors for success, to plan for and restore youth's role and position as active energy citizens in remote and Indigenous communities. Notwithstanding the recognized importance of youth engagement in energy futures [90,91], and the increased in international and national youth energy programs and opportunities [64,65,92], experience from Indigenous communities in our study region reflect a critical gap at the local level for mobilizing Indigenous youth as energy citizens. Indigenous youth participating in our study indicated a strong interest in engaging in local energy and broader community initiatives but noted that existing opportunities are deficient and non-empowering with their voices sometimes overlooked, and with few formal mechanisms to support engagement in CRE initiatives or community governance in ways that youth perceived to be impactful. This is not unique to our study communities [48] or to Indigenous youth. Similar challenges constraining youth participation in local energy governance and decision-making have been identified by scholars in other regions, including among Indigenous and non-Indigenous rural communities, and in both the renewable and non-renewable energy sectors [66,89]. Youth are often absent from local CRE initiatives despite being significantly affected – a pattern reflected also in the challenges we faced in recruiting youth to participate in our study. Without intentional, organized leadership efforts to engage youth in CRE initiatives, youth in rural and remote regions may lack the capacity and voice to engage in energy transition. The risks are high, resulting in a community energy future that may not align with local values or even future project dysfunction and failure [94].

4.2. Alignment with youth value expectations of CRE

Second, efforts to engage youth in local energy transitions must align with what youth see as the value added of CRE projects. Prevailing literature predominately addresses the climate agenda as the key driver for youth engagement in energy transitions [39], but results across our study communities underscore the significance that Indigenous youth place on more immediate and localized energy values. Lessons from PBCN and Frog Lake communities indicate that the lead values identified by Indigenous youth, and other community members, are not driven by climate change but rather the broader social and economic opportunities that CRE is expected to generate [9,48]. This includes new local employment opportunities, building new technical skills to support employment with industry and governments, and generating new income (or freeing-up existing resources) to invest in community development. Study participants emphasized the local and tangible benefits of CRE projects that immediately impact day-to-day lives. Ensuring that such benefits are realized, of course, requires more concerted efforts to mobilize youth as energy citizens – as noted above. Our findings illustrate the need to better understand what drives Indigenous youth energy citizenship at the *local* scale, which may not align with assumptions in global scholarship regarding the value(s) youth place on CRE. Further empirical research is needed to explore the different values and priorities that frame energy transition in different contexts, especially in the North and among the Indigenous youth population. To do this, scholars need to expand their focus beyond the climate agenda and consider the diverse motivations behind energy transition across various regions, cultures and communities, particularly marginalized or underrepresented communities and groups within those communities [39]. Doing so will enable a more inclusive energy transition through comprehensive policy frameworks and programs that pursue CRE opportunities that more closely align with the values, challenges, and opportunities of Indigenous youth living in remote communities.

4.3. Addressing Indigenous youth's underlying social challenges

Results from our research suggest that to empower Indigenous youth to become effective energy citizens in their communities and unlock their potential, it is essential to concomitantly address more deeply rooted social struggles. There is a growing scholarship on the historical marginalization of Indigenous communities in the context of energy infrastructure, energy transition, and as energy citizens [31]. These challenges can also significantly hinder youth from acting as enablers of energy innovation and transition in their community. Yet, the energy scholarship research that does focus on local youth engagement in CRE seems to overlook the enduring social challenges among many rural and remote Indigenous communities, especially across the North. Helping to mitigate the mental health crises that disproportionately impact Indigenous youth or reconciling feelings of alienation and cultural disconnect [99] may not be completely beyond the scope of energy citizenship. There is an opportunity through CRE research and programming to help youth to (re)connect to their cultures, values and traditions through "cultural treatment" [52] or "healing traditions" [99], which may involve school and community on-the-land learning about renewable energy resources and intergenerational dialogues to foster cultural revitalization and rebuilding trust and identity. Such initiatives are emerging in one of the communities engaged in this study, but they require further collaborative research, investment in land-based energy education, and capacity building.

4.4. Connecting youth engagement in CRE with opportunities to embrace Indigenous culture

Finally, and building on the above, meaningful youth engagement in CRE in rural and remote Indigenous communities must reflect Indigenous values and culture. Our study findings emphasize the importance of educational programming to facilitate youth engagement in CRE. Whilst the role of education in enabling and empowering youth in the energy transition is well noted in current scholarship [57,100,101], a salient observation that emerged from this research is the centrality of Indigenous values in shaping youth energy education and thus the communities' current and future CRE aspirations. That said, our study also revealed a notable lack of these values in the limited youth educational programmes currently available in the study communities. Similar findings are reported by McMaster et al. [48] in the context of remote, off-grid Gwich'in communities in Canada's Northwest Territories. Educational programs must support technical learning about energy, but this learning must be rooted in cultural values. Such cultural-based education would ensure youth connection to their communities, thus helping to increase youth engagement in CRE initiatives and, ultimately improving the viability and resiliency of CRE projects [102,103]. The southern values that shape CRE may not align well with northern communities [104] and applying southern concepts such as energy security in the north may fail to address energy challenges. Arguably, this also applies to energy literacy. The development of critical energy literacy among youth is essential for CRE transition. Numerous scholars highlight the importance of integrating Indigenous knowledge and values in transitions research, but also indicate the persistent underrepresentation of these perspectives within energy transitions scholarship [31,105,106] – this extends to research focused on youth energy education, whereby the emphasis is on testing youth's knowledge and understanding [39] rather than on how to develop and implement culturally appropriate energy education.

5. Conclusion

This research set out to examine the nature and role of youth energy citizenship in CRE in northern and Indigenous communities, specifically how youth are involved energy transitions and the barriers and opportunities to their engagement. While scholars highlight the increasing importance of youth in energy transitions, especially in the international arena and driven by climate agendas [39], this research revealed significant gaps in the context of youth engagement in CRE in northern and remote Indigenous communities. The communities in our study region are at the very early stages of CRE planning, with projects yet to be developed; however, results indicated limited youth engagement in helping drive or steer energy transitions. Youth are critical actors in the global energy transition, but ensuring youth engagement in northern and Indigenous communities requires intentionality, specifically formal means and opportunities for youth to engage on energy and broader community governance; it requires a greater understanding of the social and economic value that youth place on CRE, beyond longer-term climate goals; it requires addressing many of the underlying social challenges that Indigenous youth are facing in northern communities, and seeking opportunities for CRE engagement to help bridge some of those challenges; and it require a more concerted effort to connect youth engagement in CRE education and training with land-based Indigenous values and culture. Specifically, our research points toward four recommendations that policy makers and political leaders within Indigenous communities may consider:

- First, identify and implement programming opportunities for youth engagement in CRE at the *local* level that enable youth to participate active energy citizens.
- Second, create youth employment opportunities, through CRE projects, especially those that impact day-to-day life, that build new technical skills to support employment with industry and government and that generate new income to invest in community development.
- Third, align CRE programming to help youth to (re)connect to their cultures, values and traditions such as on-the-land learning about renewable energy resources and intergenerational dialogues to foster cultural revitalization and rebuilding trust and identity.
- Fourth, establish educational programs that support technical learning about energy, rooted in cultural values to strengthen youth connection to their communities, increase youth engagement in CRE initiatives and, ultimately, improve the viability and resiliency of CRE projects.

In conclusion, CRE provides a promising avenue to achieve energy security for Indigenous and remote communities, but the success of such projects hinges on the meaningful engagement of youth energy citizens. Future research and policy development must prioritize the reconceptualization of empowering Indigenous youth as energy citizens through addressing their challenges and enablers as active actors in CRE across diverse regions.

CRediT authorship contribution statement

Arwa Jaradat: Methodology, Formal analysis, Conceptualization, Writing – original draft. **Bram Noble:** Validation, Supervision, Project administration, Funding acquisition, Conceptualization, Writing – review & editing. **Greg Poelzer:** Validation, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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